Atomic Voice Base

SKU:A149





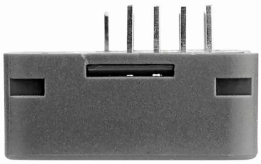














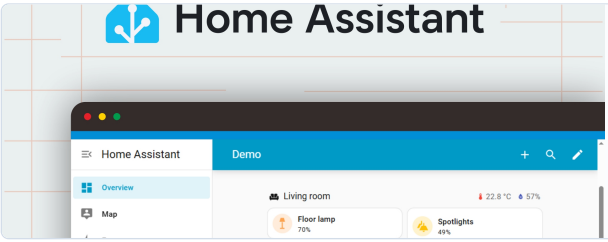




Description

Atomic Voice Base is a **voice recognition** base designed specifically for the M5 Atom series hosts. It integrates the ES8311 mono audio decoder, MEMS microphone, and NS4150B power amplifier. The ES8311 provides 24-bit audio resolution and a sampling rate of 16KHz to 64KHz, supporting I2S communication and high-fidelity audio processing. The MEMS **microphone** with high signal-to-noise ratio and digital output is suitable for automatic voice recognition, while the built-in **speaker** ensures efficient audio output. The device supports **full-duplex** communication, allowing simultaneous sound transmission and reception, enhancing interactive functions such as voice recognition, wake-up, and recording playback, making it ideal for applications in smart homes and education.

Tutorial



Home Assistant

This tutorial introduces how to combine AtomS3R + Atomic Voice Base to integrate voice assistant functions into Home Assistant.

Features

- Voice recognition
- Mono audio codec
- MEMS microphone
- Class D amplifier
- Speaker

- Development platforms
  - UiFlow1
  - UiFlow2
  - Arduino IDE

Includes

- 1 x Atomic Voice Base

Applications

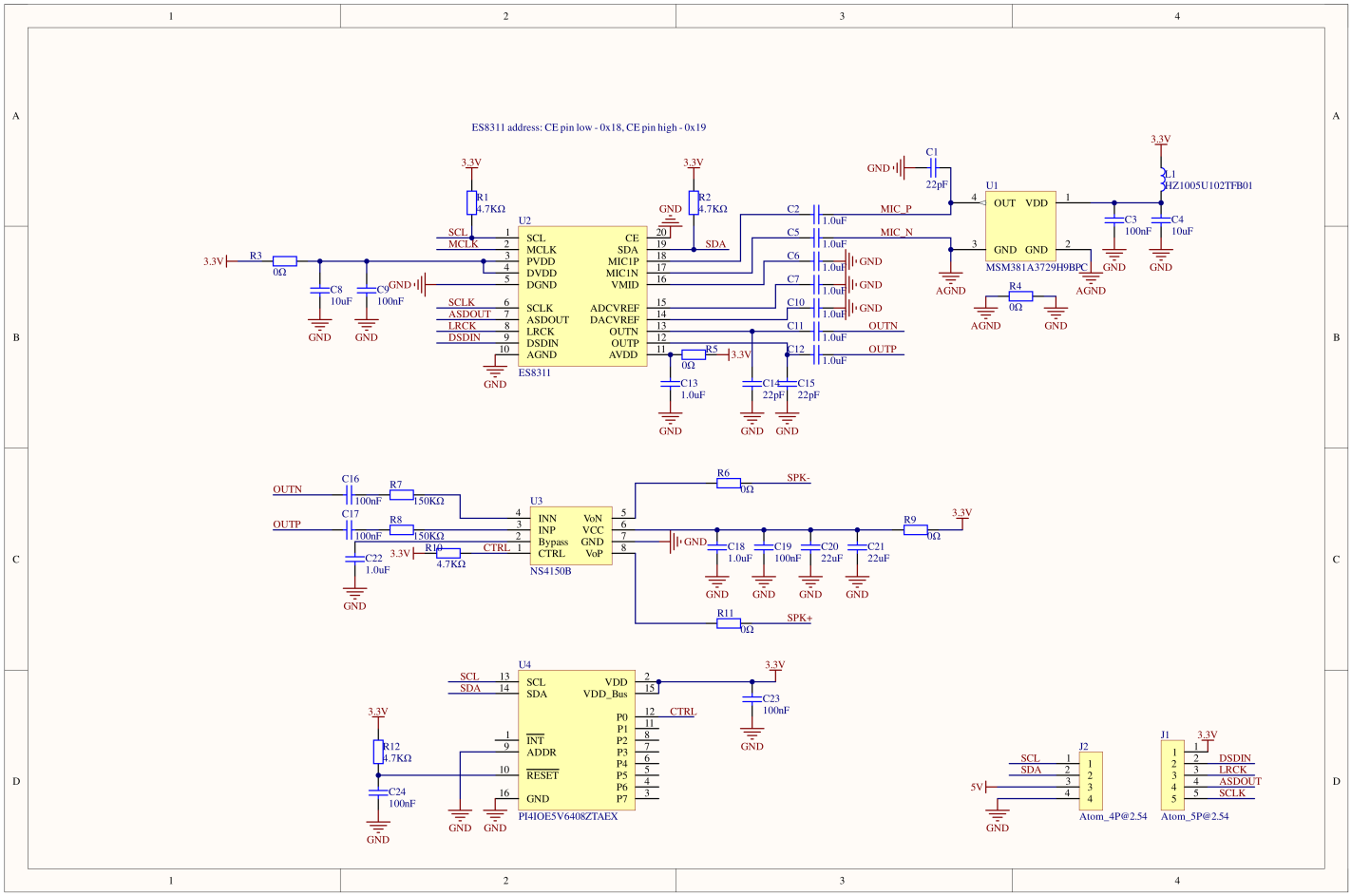
- AI voice recognition interaction systems
- Smart home control

Specifications

Specification	Parameters
Audio Codec	ES8311: 24-bit resolution, I2S protocol
MEMS Microphone	MSM381A3729H9BPC, SNR: ≥65 dB
Power Amplifier	NS4150B: Class D power amplifier
Speaker	2014 cavity speaker: 1W@8Ω
Standby Current	DC3.3V/3.36mA
Operating Current	DC3.3V/14.84mA
Operating Temperature	0 ~ 40°C
Product Size	24.0 x 24.0 x 14.1mm
Product Weight	6.5g
Package Size	85.0 x 65.5 x 15.0mm
Gross Weight	12.5g

Schematics

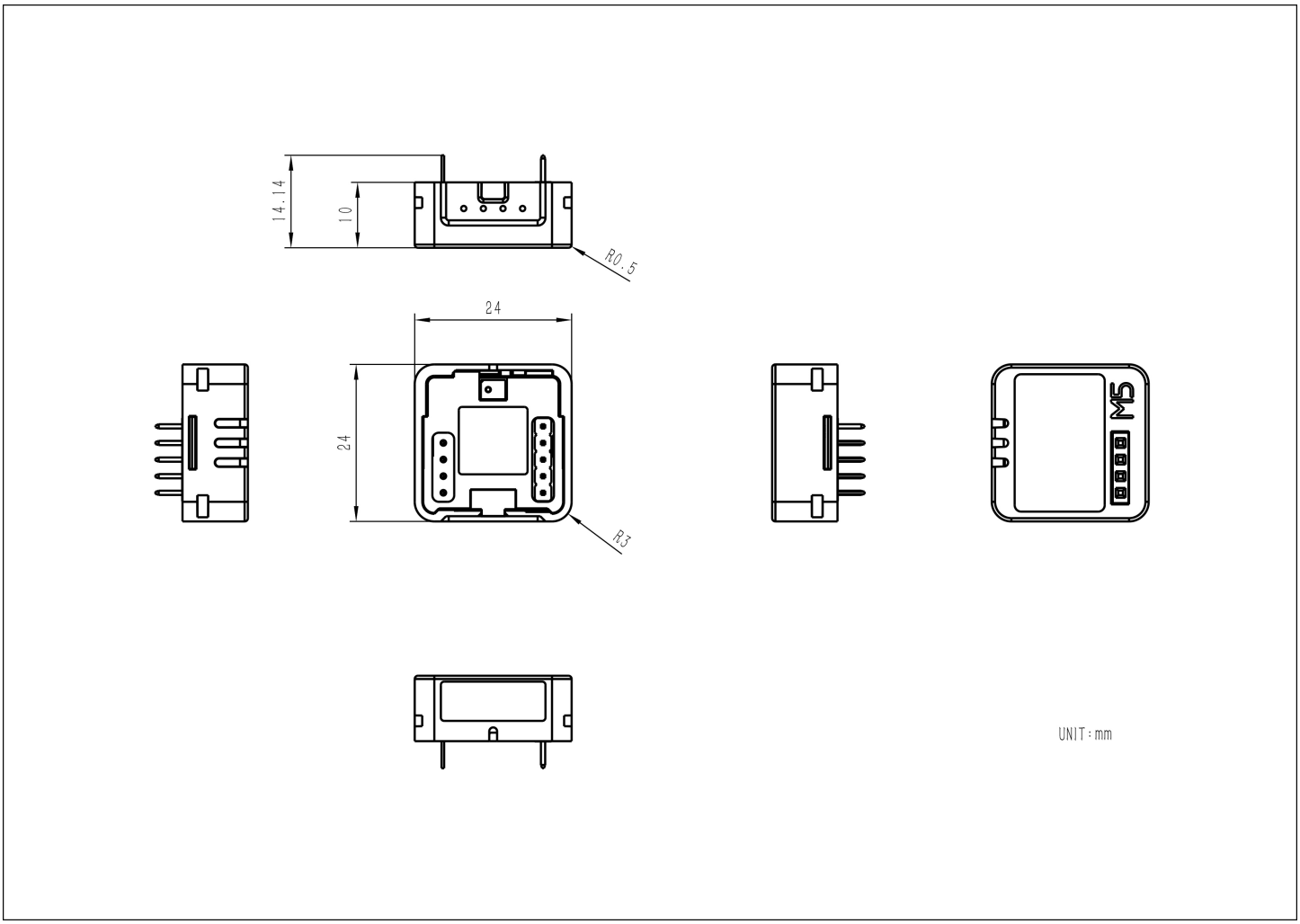
- Atomic Voice Base Schematic PDF



PinMap

PIN	LEFT	RIGHT	PIN
		1	3V3
SCL	2	3	DIN
SDA	4	5	LRCK
5V	6	7	DOUT
GND	8	9	SCK

Model Size



Datasheets

- ES8311
- MEMS MIC
- NS4150B

Softwares

MIC Sampling Rate

The Atomic Voice Base microphone supports a sampling rate range of 16KHz-64KHz

Quick Start

- OpenAI Voice Assistant For AtomS3R + Atomic Voice Base Tutorial
- Atomic Voice Base XiaoZhi Voice Assistant
- AtomS3R + Atomic Voice Base Voice Assistant

Arduino

- Atomic Voice Base Arduino Library

UiFlow1

coming soon

UiFlow2

- Atomic Voice Base UiFlow2 Docs

ESP-IDF

- Open RealtimeAPI Embedded SDK

Video

- Atomic Voice Base Product Introduction and Case Demonstration

Atomic Echo Base.mp4

- OpenAI Voice Assistant For AtomS3R + Atomic Voice Base

OpenAI Voice Assistant For AtomS3R.mp4

Product Comparison

Product Comparison Table		
	Atomic Voice Base	**Atom Voice**
Solution	Base (No SoC)	Built-in SoC (ESP32-PICO-D4)
MIC	MSM381A3729H9BPC	SPM1423
Amplifier	NS4150B@Class D amplifier	NS4150B@Class D amplifier
Speaker	1W@8Ω	0.5W@8Ω
Audio Sampling Rate	16KHz-64KHz	8kHz-96kHz
Programmable RGB/Button	/	SK6812
Communication Mode	Full-duplex	Half-duplex